



PROGRAMMING WITH PYTHON

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Data Ingestion and Multiprocessing

Data Ingestion

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PROGRAMMING WITH PYTHON

Data Ingestion

- Data ingestion is accessing and transporting data from a source or sources to a target.
- Before data can be analyzed, visualized or modeled it must be gathered from various sources and transformed into a usable format.
- Handle data from a range of sources, including flat files, spreadsheets, databases and APIs.



PROGRAMMING WITH PYTHON

Connecting Python to Database

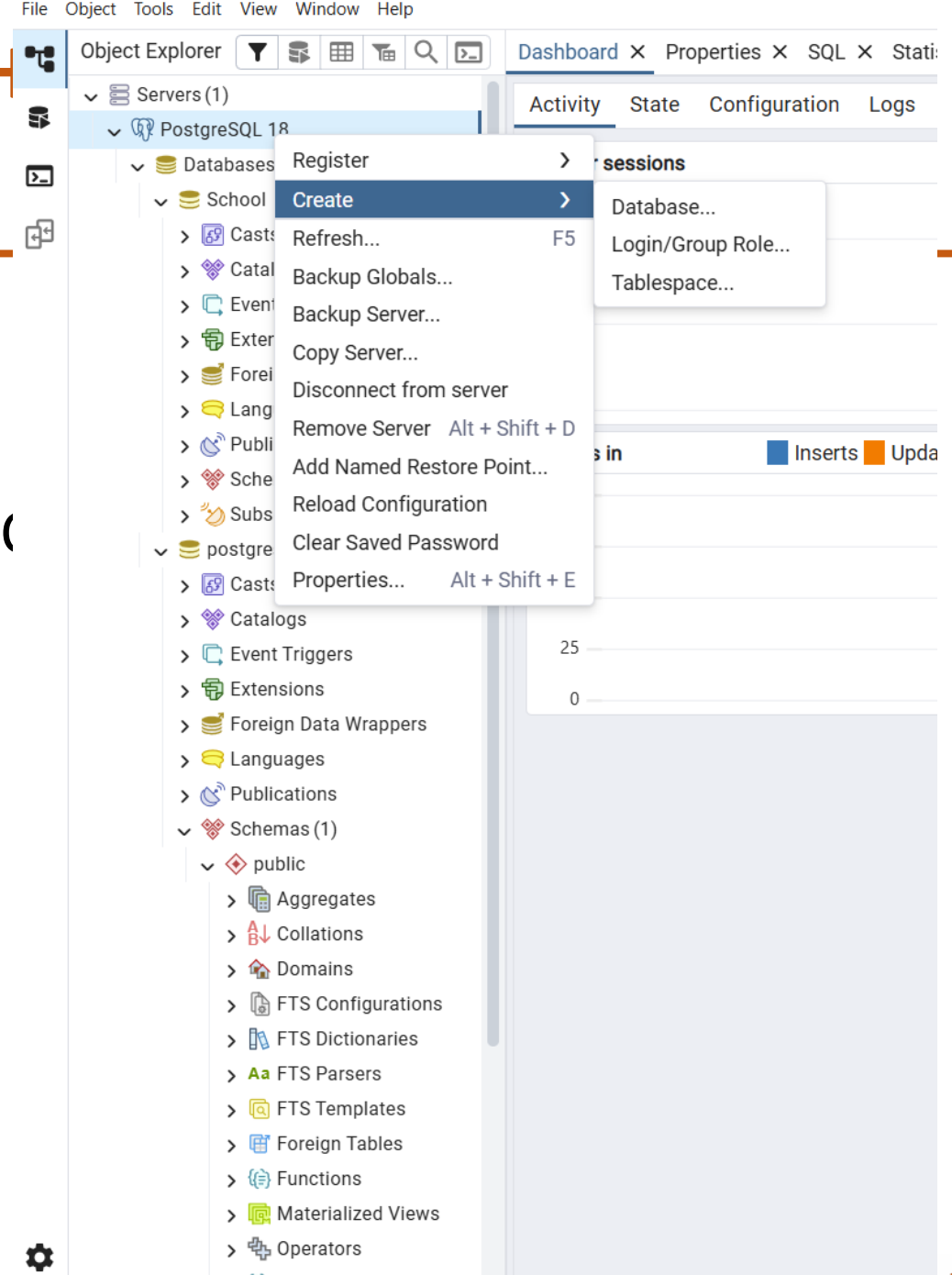
- Postgres SQL should be installed.
- In the IDE type **pip install psycopg2**.
- Run pgAdmin 4.



PROGRAMMING WITH PYTHON

Connecting Python to Database

- Create the database in pgAdmin 4.
- Right click on PostgreSQL 18 and click on (





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Connecting Python to Database

```
import psycopg2
# Connect to the School database
conn = psycopg2.connect(
    dbname="School",
    #The database name
    user="postgres",
    #User name required to authenticate.
    password="Pesu@123",
    #Password used to authenticate.
    host="localhost"
    #Database server address
)
if conn:
    print("Connection to the PostgreSQL established successfully.")
else:
    print("Connection to the PostgreSQL encountered and error.")
```

Connection to the PostgreSQL established successfully.



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Connecting Python to Database

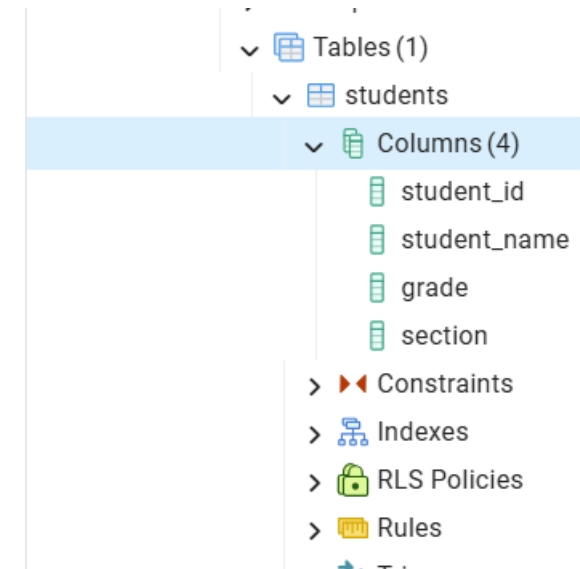
- Create a table.

```
# Cursor is opened
cur = conn.cursor()

# Table is created
cur.execute("""
    CREATE TABLE students(
        student_id SERIAL PRIMARY KEY,
        student_name VARCHAR(80) NOT NULL,
        grade VARCHAR(10) NOT NULL,
        section VARCHAR(5) NOT NULL
    );
""")

# Changes are committed
conn.commit()

# Cursor and connection are closed
cur.close()
conn.close()
```





PROGRAMMING WITH PYTHON

Connecting Python to Database

- Insert values

```
import psycopg2

conn = psycopg2.connect(
    dbname="School",
    user="postgres",
    password="Pesu@123",
    host="localhost",
    port="5432"
)

cur = conn.cursor()
# Insert rows into students table
cur.execute("INSERT INTO students(student_name, grade, section) VALUES('Amora', 'Grade 10', 'A')")
cur.execute("INSERT INTO students(student_name, grade, section) VALUES('Beast', 'Grade 9', 'B')")
cur.execute("INSERT INTO students(student_name, grade, section) VALUES('Captain America', 'Grade 8', 'C')")
cur.execute("INSERT INTO students(student_name, grade, section) VALUES('Destiny', 'Grade 10', 'B')")
cur.execute("INSERT INTO students(student_name, grade, section) VALUES('Elektra ', 'Grade 7', 'A')")

conn.commit()
cur.close()
conn.close()
```




PROGRAMMING WITH PYTHON

Connecting Python to Database

- Select Values

```
import psycopg2

conn = psycopg2.connect(
    dbname="School",
    user="postgres",
    password="Pesu@123",
    host="localhost",
    port="5432"
)
cur = conn.cursor()
cur.execute('SELECT * FROM students;')
rows = cur.fetchall()
conn.commit()
conn.close()
for row in rows:
    print(row)
```

```
(1, 'Amora', 'Grade 10', 'A')
(2, 'Beast', 'Grade 9', 'B')
(3, 'Captain America', 'Grade 8', 'C')
(4, 'Destiny', 'Grade 10', 'B')
(5, 'Elektra ', 'Grade 7', 'A')
```



PROGRAMMING WITH PYTHON

Connection with Any Database

- SQLAlchemy allows users to connect databases using Python language, run SQL queries using object-based programming and streamline the workflow.

```
import sqlalchemy as db
engine = db.create_engine(
    "postgresql://postgres:Pesu%40123@localhost:5432/School"
)
with engine.connect() as conn:
    result = conn.execute(db.text("SELECT * FROM students"))
    print(result.fetchall())
```

```
[(1, 'Amora', 'Grade 10', 'A'), (2, 'Beast', 'Grade 9', 'B'), (3, 'Captain America', 'Grade 8', 'C'), (4, 'Destiny', 'Grade 10', 'B'), (5, 'Elektra ', 'Grade 7', 'A')]
```



THANK YOU

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